NHẬP MÔN BIG DATA

Khoa CNTT - HUIT

Binary Digit

Bội số của byte theo IEC 60027-2		
Tiền tố SI		
Tên	**Ký hiệu**	**Bội số**
kilobyte	kB	10^3 (hay 2^{10})
megabyte	MB	10^6 (hay 2^{20})
gigabyte	GB	10^9 (hay 2^{30})
terabyte	TB	10^{12} (or 2^{40})
petabyte	PB	10^{15} (hay 2^{50})
exabyte	EB	10^{18} (hay 2^{60})
zettabyte	ZB	10^{21} (hay 2^{70})
yottabyte	YB	10^{24} (hay 2^{80})

The current state of data



The growth of data

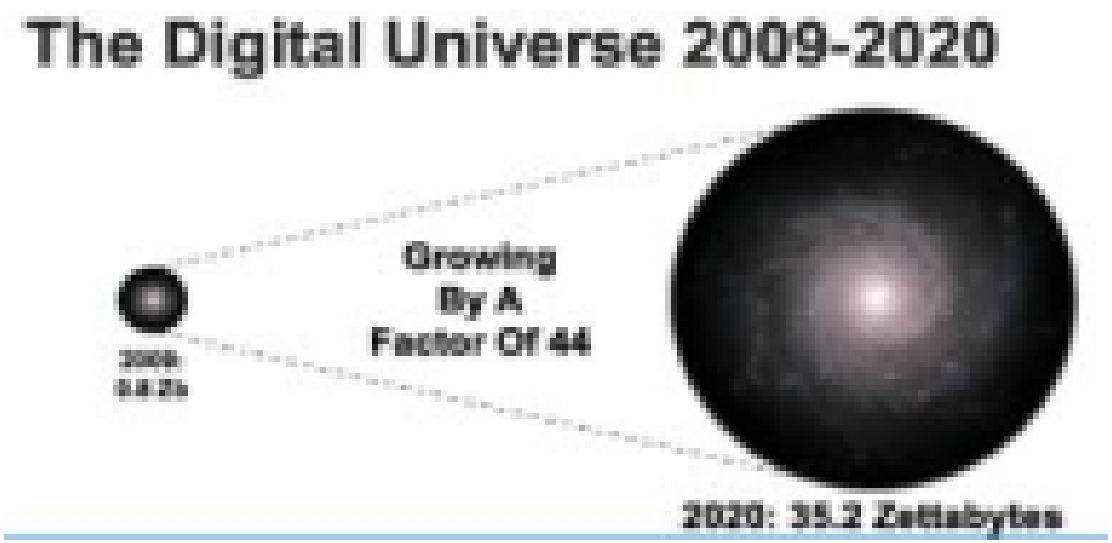


- Every day we create
  2,500,000, 000,000, 000,000
  (2.5 quintillion)

This would fill 10 million blu-ray discs

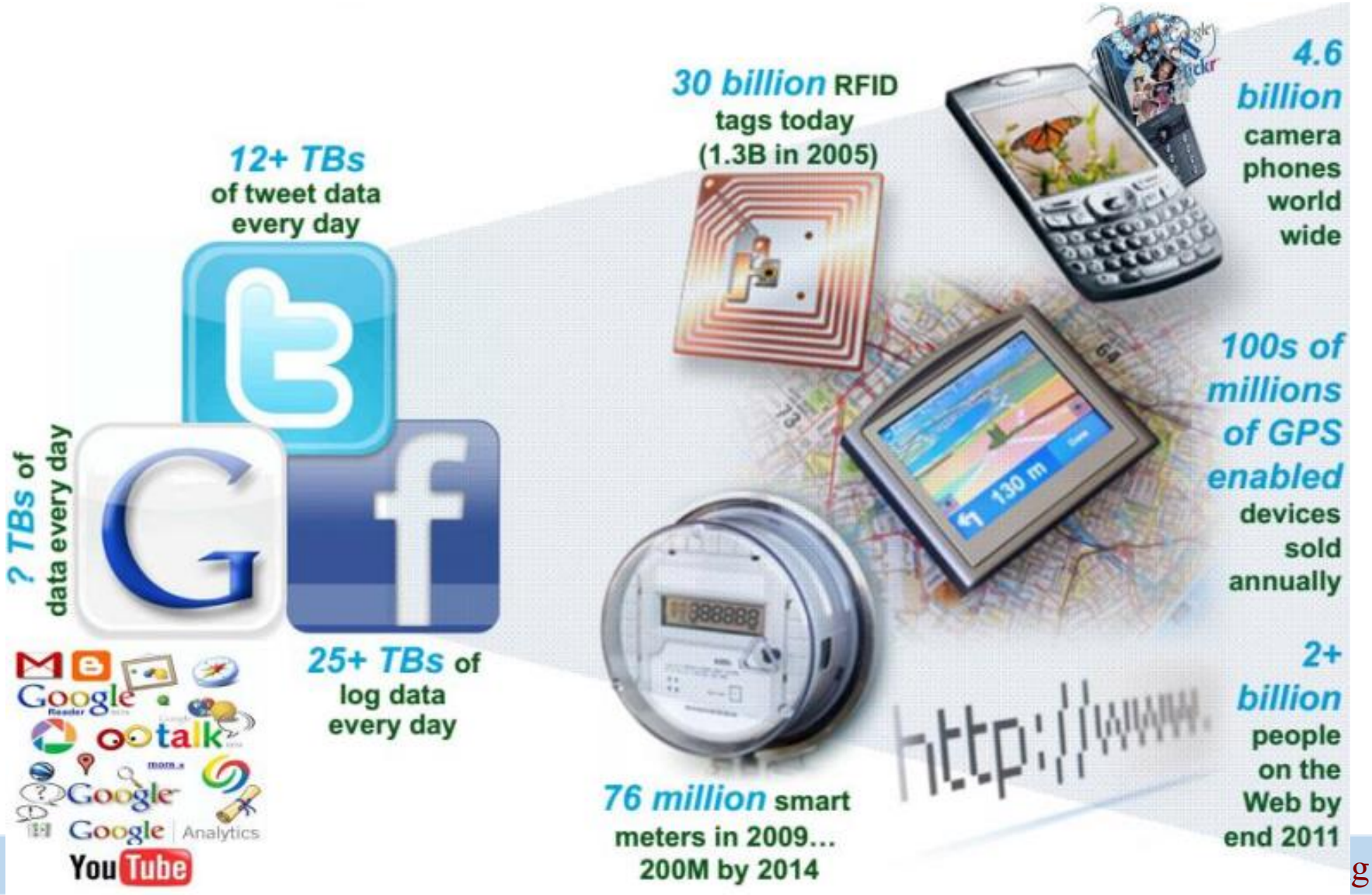
The height of which stacked, would measure the height of 4 Eiffel Towers

Data volume (scale)



- Data Volume
  + 44 x increase from 2009 to 2020
  + From 0.8 zettabytes to 35zb
- Data volume is increasing exponentially.

The social layer in an instrumented interconnected world



The Large Hadron Collider (LHC) generates **30 Petabytes** a year



The Earthscope project

It is largest science project in the world.



+ Designed to track North America's Geological.

+ This observatory records data over 3.8 million square miles, 67 terabytes of data

Other statistics of data growth

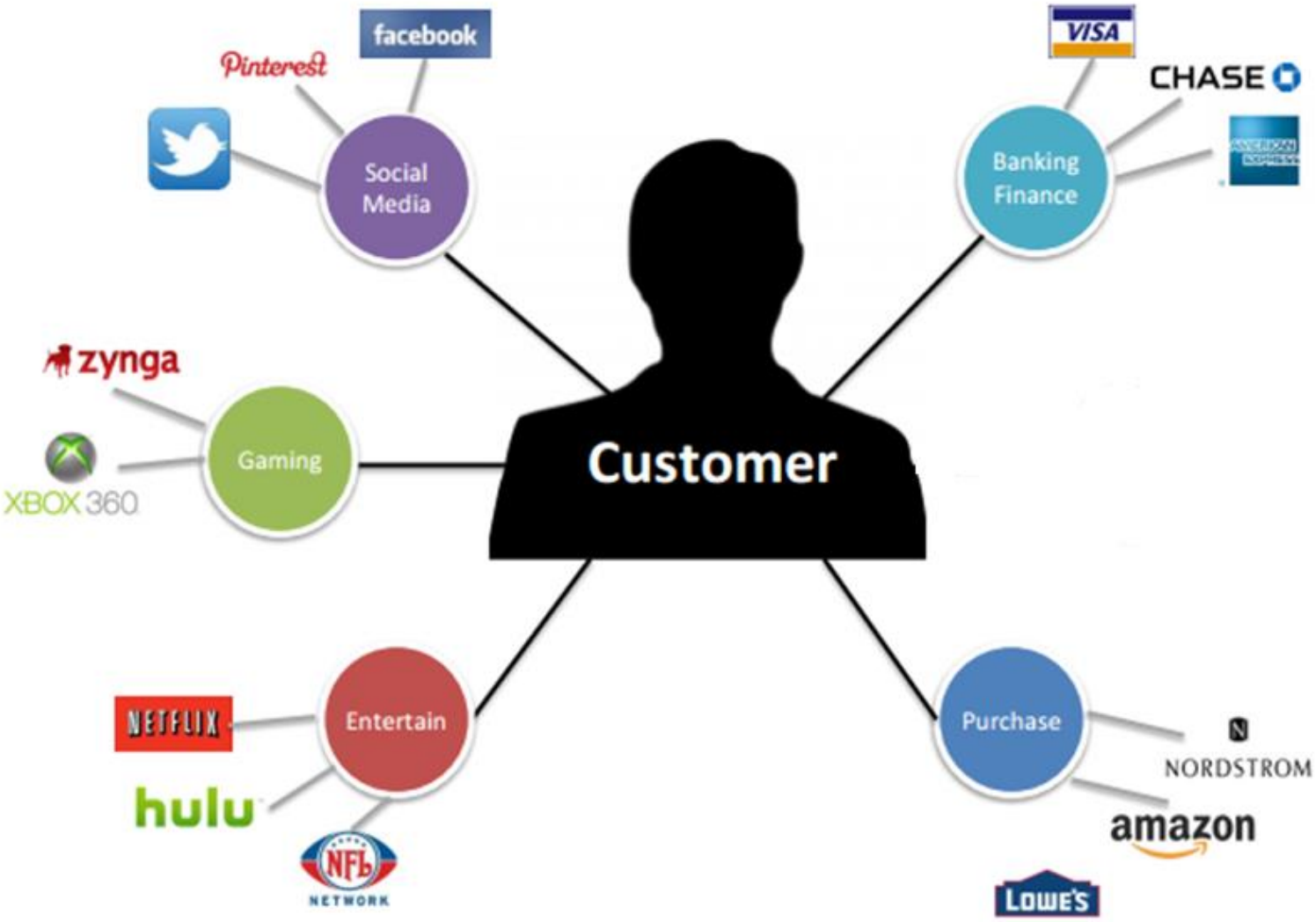
- The New York Stock Exchange generates about 4-5 terabytes of data per day.
- Facebook hosts more than 240 billion photos, growing at 7 petabytes per month.
- Ancestry.com, the genealogy site, stores around 10 petabytes of data.
- The Internet Archive stores around 18.5 petabytes of data

Data complexity (Variety)

- Various formats, types, and structures
- Text, numerical, images, audio, video,
- Sequences, time series, sensor data, user logs, social media data.
- A single application can be generating/collecting many types of data.

To extract knowledge → all these types of data need to linked together

A single view to the customer



Data speed (Velocity)

- Data is begin generated fast and need to be processed fast.
  + Online Data Analytics
  + Late decisions → missing opportunities

Real-time/Fast data



Social media and networks
(all of us are generating data)



Scientific instruments
(collecting all sorts of data)



Mobile devices
(tracking all objects all the time)



Sensor technology and networks
(measuring all kinds of data)

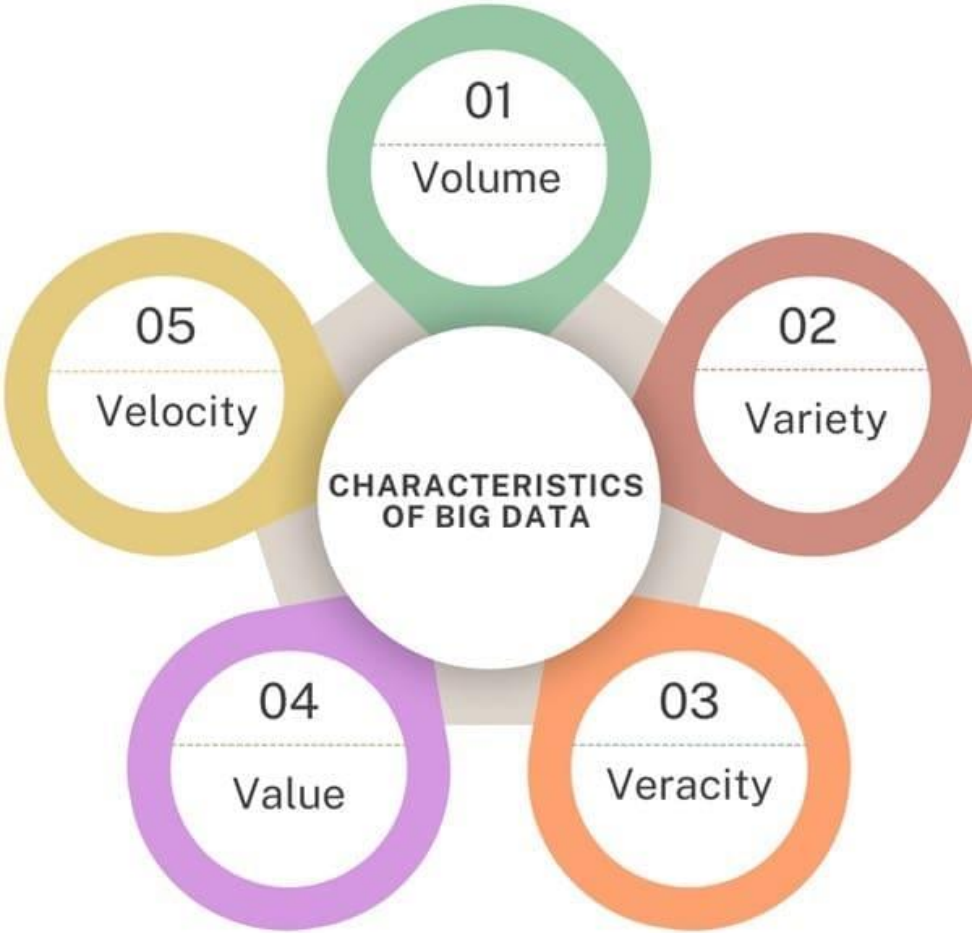
An overview of Big Data



Concept of “Big Data”

Big data is a term for data sets that are so large or complex that traditional data processing application software are inadequate to deal with them.

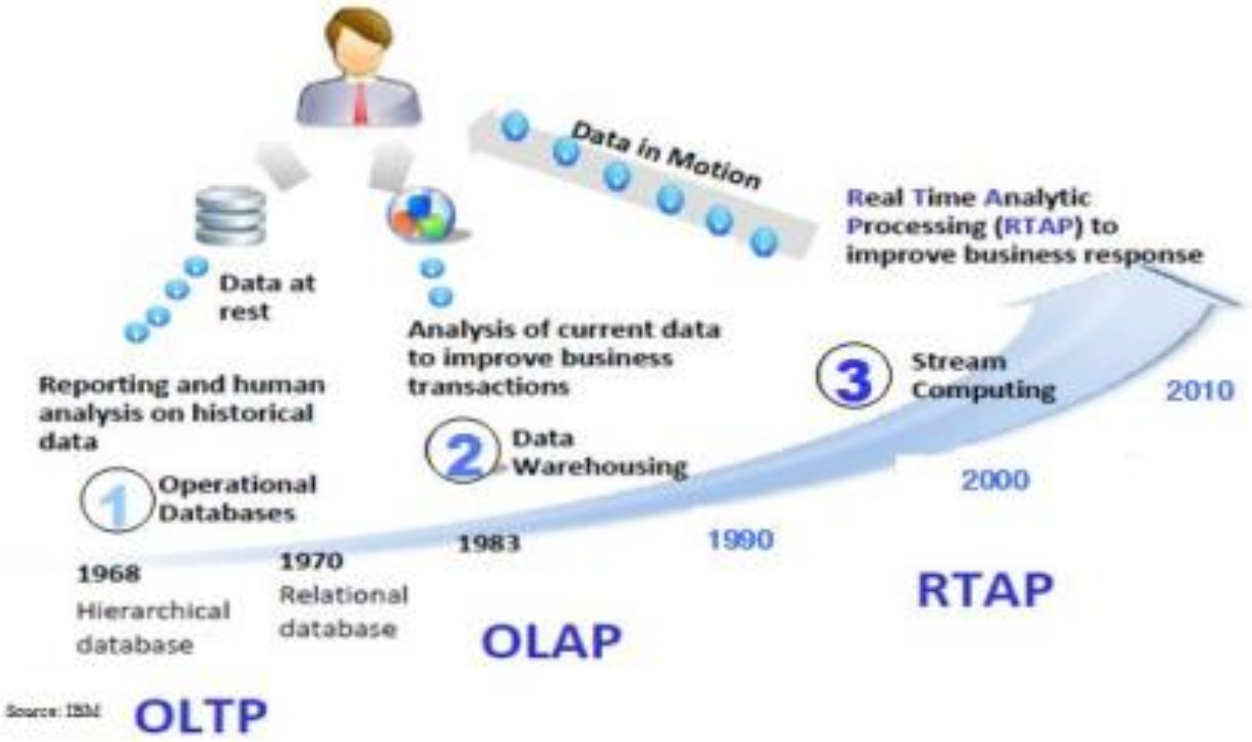
Characteristics of Big Data



Characteristics of Big Data

- **Volume**: The massive size and sheer quantity of data, often measured in terabytes, petabytes, or exabytes.
- **Velocity**: The high speed at which new data is created, collected, and streamed in real-time.
- **Variety**: The different types and formats of data, including structured tables, semi-structured files, and unstructured text, audio, or video.
- **Veracity**: The trustworthiness, quality, and accuracy of the collected data.
- **Value:** The practical business worth or actionable insights gained after analyzing the raw data.

Harnessing Big Data



- OLTP: Online Transaction Processing (DBMSs)
- OLAP: Online Analytical Processing (Data Warehousing)
- RTAP: Real-Time Analytics Processing (Big Data Architecture & technology)

Who is generating Big Data?

- **Old model:** Few companies are generating data, all others are consuming data.



- **New model:** All of us are generating data, and all of us are consuming data



Challenges in handling Big Data

- New architecture, algorithms, techniques are needed
- Lack of experts in using the new technology and dealing with big data
- Efficiently storing and accessing large amounts of data is difficult
- Managing metadata for semi-structured and unstructured data, generated by diverse sources, is a convoluted problem

Techniques towards Big Data

- Parallel
- Huge Data Volumes Storage
- Data Distribution
- High-Speed Networks
- High-Performance Computing
- Task and Thread Management
- Data Mining and Analytics
- Machine Learning
- Data Visualization

Big Data platform

- Technologies applied to Big Data should include

